

Eric Du

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<https://ericsportfolio1.netlify.app/>

Education

University of Washington, Paul G. Allen School of Computer Science

B.S. in Computer Science

• **Relevant coursework:** Foundations of Computer Science, Object Oriented Design, Data Structure and Algorithms, System Design, Linear Algebra, Software Design

June 2027(Expected)

Seattle, WA

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Work Experience

Seattle Software Developers

Full Stack Engineer

- **Led full-stack Development** and deployment of **10+** scalable web application using the **MERN** stack (**MongoDB, Express.js, React.js, Node.js**).
- Led a development team using **Agile methodologies** and **Git** for version control to build an enterprise solution integrated with **SAP**. Utilized **AWS Cognito** for authentication and **AWS EC2** for scalable deployment, resulting in a **70%** improvement in loading time.
- Trained and deployed **AI language models** with **Amazon Lex**, doubling response speed and enhancing user interaction performance.

June 2025 - Sep 2025

Seattle, WA

Projects

SeaHoop

- Designed and deployed **geofencing, bounding box search, and viewport-based data loading algorithms**, to efficiently fetch and display nearby court activities using the **Google Map API's**.
- Utilized **Firebase** as data storage, **Firebase Authentication**, and **Swift UI** for IOS development.
- Implemented **real-time location-based updates** and **predictive traffic modeling** to estimate future court crowd levels using historical usage patterns and time-series analysis.
- Led UI/UX design and community outreach, growing the user base to **4000+ active users in 3 months**

AWS Cognito Intelligent Banking System

- Developed a full-stack **Node.js/Express** backend with **OpenID Connect, Session Management**, and **RESTful APIs** for seamless frontend-backend communication.
- Implemented **AWS Cognito** for highest authentication security, optimized log in speed by **40%**.
- Trained and integrated **Deepseek-V3** into a custom **AI chatbot**, enabling accurate and context-aware responses to banking-related queries with **enhanced natural language** understanding.
- Utilized modern authentication **protocols (OIDC/OAuth2)** to protect sensitive financial data and streamline user onboarding.

Technical Skills

Languages: Java, Python, JavaScript, Swift, C/C++, ABAP, HTML, CSS

Developer Tools: Git, VS Code, IntelliJ, JUnit, Agile /Scrum

Technology/Frameworks: PyTorch, TensorFlow, React.js, Node.js, Express.js, MongoDB, Amazon Web Services (EC2, Cognito, Lex), SAP, Natural Language Processing (NLP), Machine Learning, REST APIs, Firebase.